Hunting a Counterexample to Agrawal's Conjecture: Background and Strategy

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Audit notice (22 October 2026). The two odd-cardinality cases and the subset-product modulus definitions below have been corrected. Other feasibility statements remain historical estimates. The same-name PDF beside this source is historical; the rebuilt audit version is in `doc/audit-pdfs/`. See `doc/integration-review-status.md` for current review status.

1 The conjecture and its status

For n, r coprime, write $T(-1, n, r)$ for the congruence

$$(X-1)^n \equiv X^n - 1 \pmod{n,\ X^r - 1}. \tag{1}$$

Conjecture 1 (Agrawal; Bhattacharjee–Pandey 2001)**.** *If r is a prime not dividing n and (1) holds, then n is prime or $n^2 \equiv 1 \pmod r$.*

If true, a single congruence (1) at one $r = O(\log n)$ certifies primality, giving a deterministic $\tilde{O}(\log^3 n)$ test — versus $\tilde{O}(\log^6 n)$ for the best proven variant (Lenstra–Pomerance Gaussian periods [10]). The status is peculiar: Lenstra and Pomerance [2] gave a heuristic predicting $\gg x^{1-\varepsilon}$ counterexamples below x, yet none is known; exhaustive search (Primaboinca, 2010–2020) cleared $n < 10^{17}$, and folklore places the smallest counterexample at hundreds of digits. Popovych [3] proposed the repair of adding $(X+2)^n \equiv X^n + 2$ as a second hypothesis; that variant is untouched by any heuristic or search.

Our angle: abandon the smallest counterexample. Any counterexample disproves the conjecture, the heuristic density improves with size, and the construction below is deterministic — if its arithmetic side conditions hold, the congruence (1) is guaranteed.

2 The Lenstra–Pomerance construction ($r = 5$)

Theorem 1 (Lenstra–Pomerance [2]; statement as in Váňa [4], who supplies the $k \equiv 3$ case)**.** *Let $n = p_1 \cdots p_k$ with p_i distinct primes such that*

(a) k is odd;

(b) $p_i \equiv 3 \pmod{80}$ for all i;

(c) $p_i - 1 \mid n - 1$ for all i (n is a Carmichael number);

(d) $p_i + 1 \mid n + 1$ for all i (n is a Lucas–Carmichael number).

Then $T(-1, n, 5)$ holds and $n^2 \not\equiv 1 \pmod 5$.

Why this works, in one paragraph. Since $p \equiv 3 \pmod 5$ has order 4 in $(\mathbb{Z}/5)^\times$, over $\mathbb{F}_p$ we have $X^5 - 1 = (X-1)\Phi_5(X)$ with Φ_5 irreducible, so $\mathbb{Z}[X]/(p, X^5-1) \cong \mathbb{F}_p \times \mathbb{F}_{p^4}$. The $\mathbb{F}_p$ component of (1) is trivial. In $\mathbb{F}_{p^4}$, with ζ a primitive fifth root of unity, Frobenius gives $(\zeta-1)^{p^2} = -\zeta^{-1}(\zeta-1)$, hence $\rho(p) \mid 10(p^2-1)$. Choose $j = 1$ or 3 according to k mod 4. Conditions (c),(d) give $n \equiv p^j$ modulo the odd numbers $(p-1)/2$ and $(p+1)/4$; the residue $n \equiv 3^k \equiv p^j \pmod{80}$ handles the remaining factor. These three moduli are pairwise coprime and their product is $10(p^2-1)$, so $n \equiv p^j \pmod{10(p^2-1)}$. Thus $(\zeta-1)^n = (\zeta-1)^{p^j} = \zeta^{p^j} - 1 = \zeta^n - 1$. Finally $p_i \equiv 3 \pmod 5$ and k odd give $n \equiv 3^k \in \{2, 3\} \pmod 5$, so $n^2 \not\equiv 1 \pmod 5$.

The pain is concentrated in (c)∧(d):

Lemma 1 (Side-coprimality)**.** *If n satisfies (c) and (d), then* $\gcd(p_i - 1,\, p_j + 1) \mid 2$ *for all* i, j.

Proof. An odd prime q dividing both divides $n - 1$ and $n + 1$, hence 2. □

So the odd primes supporting all the $p_i - 1$ and those supporting all the $p_i + 1$ must be *disjoint sets*. No number that is simultaneously Carmichael and Lucas–Carmichael is known (none exists below 10^{18}, by Pinch's tables [7]); Lemma 1 is the structural reason, and the reason the smallest example is expected to be enormous. Note (b) gives $v_2(p-1) = 1$, $v_2(p+1) = 2$, $5 \nmid p^2 - 1$, and for k odd the 2-adic parts of (c),(d) are automatic ($n \equiv 3 \bmod 4$); the problem lives entirely in odd moduli prime to 5.

3 Reduction to a subset-product problem

This is now a job for the Erdős-style machinery behind large-scale Carmichael constructions [5, 6]. Fix disjoint sets Q_-, Q_+ of odd primes $\neq 5$ and harvest the pool

$$\mathcal{P} \;=\; \big\{\, p \le x \;:\; p \equiv 3 \pmod{80},\ \tfrac{p-1}{2} \text{ is } Q_-\text{-smooth},\ \tfrac{p+1}{4} \text{ is } Q_+\text{-smooth} \,\big\}.$$

Put $\Lambda_- = \operatorname{lcm}\{p-1\} = 2\operatorname{lcm}\{(p-1)/2\}$ and $\Lambda_+ = \operatorname{lcm}\{p+1\} = 4\operatorname{lcm}\{(p+1)/4\}$ over a nonempty $\mathcal{P}$. Disjoint odd supports give $\gcd(\Lambda_-, \Lambda_+) = 2$, so the targets below are compatible. For any $S \subseteq \mathcal{P}$ with $|S|$ odd, $|S| \ge 3$, the number $n = \prod_{p \in S} p$ satisfies Theorem 1 provided

$$n \equiv 1 \pmod{\Lambda_-} \qquad \text{and} \qquad n \equiv -1 \pmod{\Lambda_+}, \tag{2}$$

a condition *stronger* than (c),(d). Restrict to pool elements coprime to $\Lambda_-\Lambda_+$ before taking discrete logarithms in each cyclic component of $(\mathbb{Z}/\Lambda_-)^\times \times (\mathbb{Z}/\Lambda_+)^\times$ (Pohlig–Hellman is cheap — all orders are smooth by design), (2) becomes a 0/1 subset-sum over $\bigoplus_i \mathbb{Z}/m_i$ with one extra $\mathbb{Z}/2$ component for the parity of $|S|$. Heuristically solvable once

$$2^{|\mathcal{P}|} \;\gg\; |G|, \qquad |G| \approx \varphi(\Lambda_-)\,\varphi(\Lambda_+),$$

i.e. $|\mathcal{P}| \gtrsim \log_2 \Lambda_- + \log_2 \Lambda_+$ with slack for the solver. This needs a uniformity/reachability hypothesis; the size inequality alone does not guarantee a subset attaining the target. Ballpark: a 15–30k-bit group wants a pool of 50–100k primes. Solvers, in escalating order: staged CRT à la Löh–Niebuhr [6], Wagner's k-list generalized birthday algorithm [8] (the pool-rich regime is exactly its habitat), lattice reduction only for terminal low-dimensional corrections.

Two design points. *Harvesting is constructive, not sieved*: enumerate Q_--smooth m, set $p = 2m + 1$, test $p \equiv 3 \pmod{80}$ and primality, then trial-divide $(p+1)/4$ over Q_+; this reaches $x = 10^{12+}$ easily. An *asymmetric split* looks best: $Q_+ = \{\text{odd primes} \le B\} \setminus \{5\}$ (the "random side", where $(p+1)/4$-smoothness is Dickman-governed), $Q_- \subset (B, B']$ medium primes (the "constructive side", disjointness automatic). The resulting n would have thousands of prime factors and 10^4–10^5 digits; verifying (1) directly at that size is a few core-hours, and testing Popovych's second congruence on any find is free — either we kill both conjectures or we get the first data separating them.

4 Odds, honestly

For: the double-condition pipeline appears never to have been run at scale — Váňa mined existing Carmichael tables (wrong haystack), Hegde–Devaraj [9] stopped at density heuristics, and the modern subset-sum toolkit postdates the 2003 construction. Against: Lemma 1 thins the pool — both-sided smoothness costs roughly $\rho(u)^2$ (Dickman) *before* the disjointness tax, and if $\log|G|$ grows linearly with the pool the feasibility inequality never closes. Phase 1 of `PLAN.md` measures exactly this and produces a go/no-go number before any solver is built; a quantified infeasibility result for the L–P route would itself be worth writing up.

References

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